Information-Driven Bars for Financial Machine Learning: Imbalance Bars

By Gerard Martínez (Towards Data Science)

Archival summary card for the Trading Strategy documentation collection. This is an original summary, not a reproduction of the source article. Read the full article at the link below.

A practical tutorial on imbalance bars, the most information-driven member of the López de Prado bar family, following on from tick, volume and dollar bars. The motivation from Advances in Financial Machine Learning (2018): sample more frequently when new information arrives, synchronising sampling with the arrival of informed traders so you may act before prices reach a new equilibrium. Imbalance bars apply to tick, volume or dollar data to produce tick (TIB), volume (VIB) and dollar (DIB) imbalance bars.

The construction works off signed ticks (the tick rule assigns +1/-1 to each trade by price direction): accumulate the signed imbalance trade by trade, and close a new bar when the cumulative imbalance breaches an expected threshold, at which point it resets to zero and a fresh threshold is computed. The threshold is the product of an EWMA of recent bar sizes and an EWMA of the imbalance proportion, so a run of same-signed aggressive trades trips a bar quickly while balanced two-sided flow lets it run.

The article is candid about the practical pitfalls: reference implementations disagree, TIBs are sensitive to how the expected-candle-size EWMA is defined, and the recursive threshold can be numerically finicky. A good on-ramp before using a library such as mlfinlab.

Source: https://medium.com/data-science/information-driven-bars-for-financial-machine-learning-imbalance-bars-dda9233058f0